

SIMATS ENGINEERING

CO1 - ASSESSMENT TOOL 2

ERROR ANALYSIS TASK

Name of the Student	V Sunil Kumar
Reg. No	192425292

COURSE NAME: Deep Learning

COURSE CODE: MLA0403

FACULTY : Dr.Arul Raj A.M

COURSE OUTCOME COVERED

CO 1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BTL-4)

CO1 Weightage: 15%

Assessment Tool Weightage: 30%

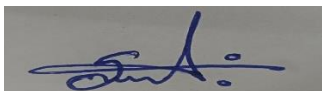
Duration: 30 minutes

Marks: 25

Code of Conduct:

I V Sunil Kumar/192425292(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

QUESTIONS:

Total Marks:25

1. Error Analysis in Maximum Likelihood Estimation (MLE)

A student builds a binary classifier using Maximum Likelihood Estimation. The likelihood function is incorrectly written as the sum of probabilities instead of the product. The model produces poor parameter estimates even with large data.

Tasks:

a) Identify the conceptual error in the likelihood formulation.

Answer :

The likelihood was incorrectly written as the sum of probabilities instead of the product of probabilities. For independent observations, the likelihood must be the product of individual probabilities.

b) Explain how this mistake affects parameter estimation.

Answer :

Using the sum instead of the product gives an incorrect likelihood function. As a result, the estimated parameters do not maximize the true likelihood, leading to inaccurate predictions even with large datasets.

C) Suggest the correct mathematical formulation and reasoning.

Answer :

The correct likelihood is:

$$L(\theta) = \prod_{i=1}^n P(x_i | \theta)$$

This product represents the joint probability of all independent observations and provides the correct parameter estimates.

c) How would using log-likelihood fix computational issues?

Answer :

The log-likelihood converts the product into a sum:

$$\log L(\theta) = \sum_{i=1}^n \log P(x_i | \theta)$$

This avoids numerical underflow, simplifies differentiation, and makes optimization easier.

2. Error Diagnosis in Perceptron Learning

While implementing a Perceptron, a student observes that the model fails to converge even after many iterations on a linearly separable dataset.

Tasks:

a) List possible implementation errors in the learning algorithm.

Answer :

- Incorrect weight update formula.
- Wrong learning rate.
- Incorrect bias update.
- Errors in prediction or activation function.
- Improper stopping condition.

b) Analyze how incorrect learning rate or weight updates affect convergence.

Answer :

A very high learning rate causes unstable updates, while a very low learning rate slows learning. Incorrect weight updates prevent the perceptron from finding the correct decision boundary.

c) Identify dataset-related issues that may cause this problem.

Answer :

- Incorrect class labels.
- Noisy or duplicate data.
- Features not properly scaled.
- Data not actually linearly separable.

d) Propose modifications to ensure convergence.

Answer :

- Use the correct weight update rule.
- Normalize feature values.
- Choose an appropriate learning rate.
- Verify that the dataset is linearly separable.
- Shuffle training data each epoch.

3. Backpropagation Gradient Error Analysis

A Multilayer Perceptron trained using Backpropagation shows increasing loss during training instead of decreasing.

Tasks:

a) Identify possible mathematical or coding errors in gradient computation.

Answer :

- Incorrect derivative calculation.
- Wrong chain rule implementation.
- Incorrect weight or bias updates.
- Wrong learning rate sign.
- Programming bugs in gradient computation.

b) Explain the role of activation functions in gradient flow.

Answer :

Activation functions introduce non-linearity. Their derivatives determine how gradients flow through the network during backpropagation.

c) Analyze how vanishing or exploding gradients affect training.

Answer :

Vanishing gradients make weight updates very small, slowing learning. Exploding gradients produce very large updates, making training unstable and increasing loss.

d) Suggest techniques to fix the issue (e.g., normalization, initialization).

Answer :

- Normalize input data.
- Use Xavier or He weight initialization.
- Apply Batch Normalization.
- Use ReLU activation.
- Apply Gradient Clipping.

4. Gradient Descent vs SGD Error Investigation

A model trained using Gradient Descent converges very slowly, while the same model with Stochastic Gradient Descent shows unstable loss fluctuations.

Tasks:

a) Explain why batch gradient descent is slow in large datasets.

Answer :

Batch Gradient Descent computes gradients using the entire dataset before each update, making each iteration computationally expensive.

b) Analyze the cause of fluctuations in SGD.

Answer :

SGD updates weights using one sample at a time. Since each sample is different, the gradient estimates are noisy, causing fluctuations in the loss.

c) Compare the error behavior of Mini-Batch Gradient Descent.

Answer :

Mini-Batch Gradient Descent uses small batches of data. It reduces the noise of SGD while being faster than Batch Gradient Descent, providing more stable convergence.

d) Recommend the best approach and justify with error trade-offs.

Answer :

Mini-Batch Gradient Descent is the preferred approach because it balances training speed, computational efficiency, and convergence stability while reducing loss fluctuations.

5. Curse of Dimensionality Error Analysis

A machine learning model built using high-dimensional data performs well on training data but poorly on test data due to overfitting, related to the Curse of Dimensionality.

Tasks:

a) Explain how high dimensionality increases model error.

Answer :

High-dimensional data increases model complexity, making it easier to fit noise instead of meaningful patterns, leading to higher prediction errors on new data. Identify signs of overfitting in this scenario.

b) Analyze why distance-based learning fails in high dimensions.

Answer :

- Very high training accuracy.
- Low test accuracy.
- Large gap between training and testing performance.
- Poor generalization to unseen data.

c) Suggest dimensionality reduction or regularization techniques to improve performance.

Answer :

In high-dimensional spaces, distances between data points become very similar. This makes it difficult for algorithms like KNN to distinguish between nearby and distant samples.

d) Suggest dimensionality reduction or regularization techniques to improve performance.

Answer:

- Apply Principal Component Analysis (PCA).
- Use Feature Selection to remove irrelevant features.
- Apply L1/L2 Regularization.
- Use Dropout in neural networks.
- Perform Cross-Validation to improve generalization.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem

Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

		explained solution			
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